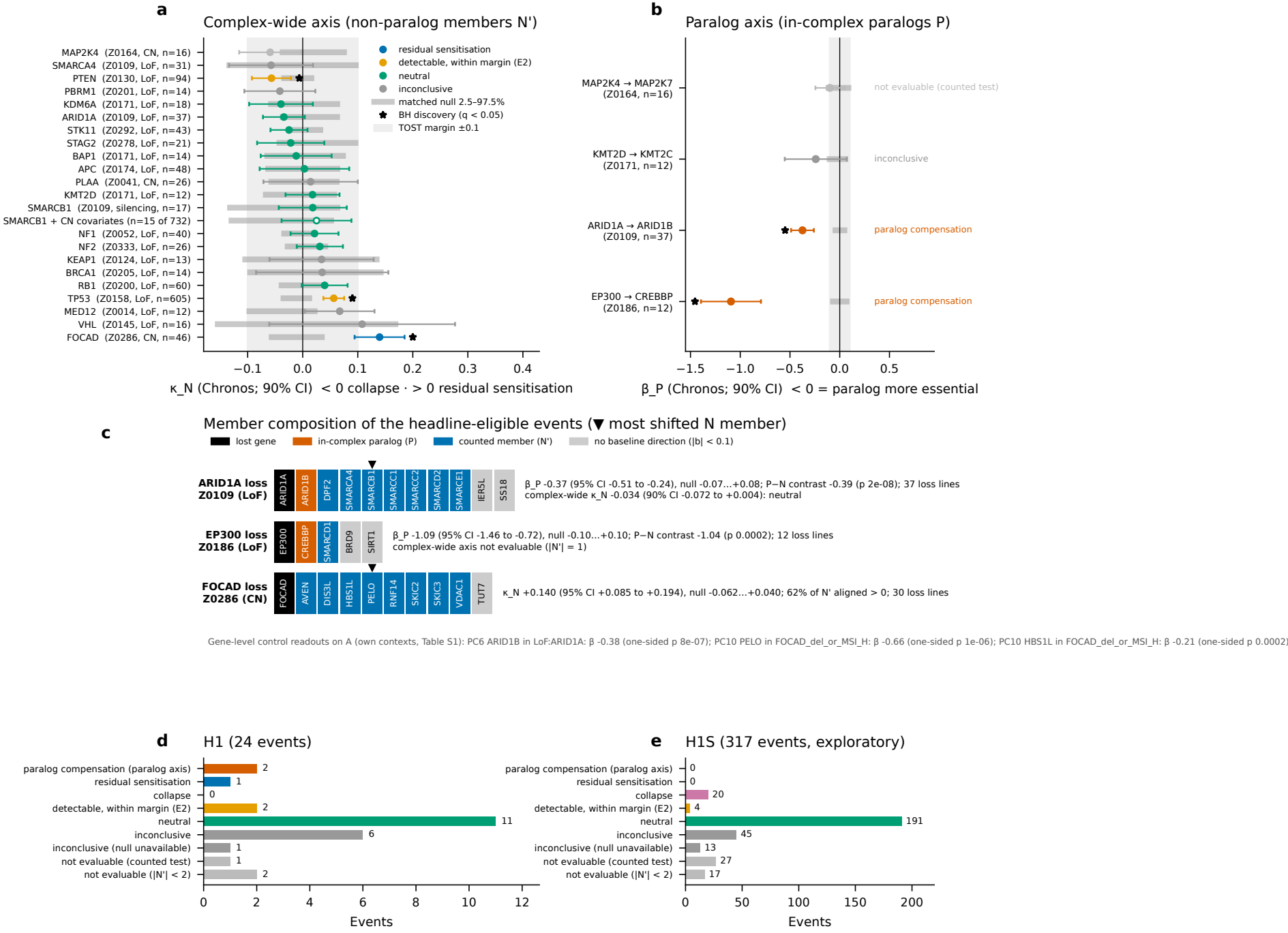
a
Complex-wide axis (non-paralog members N')
MAP2K4 (Z0164, CN, n=16)
SMARCA4 (Z0109, LoF, n=31)
PTEN (Z0130, LoF, n=94)
PBRM1 (Z0201, LoF, n=14)
KDM6A (Z0171, LoF, n=18)
ARID1A (Z0109, LoF, n=37)
STK11 (Z0292, LoF, n=43)
STAG2 (Z0278, LoF, n=21)
BAP1 (Z0171, LoF, n=14)
APC (Z0174, LoF, n=48)
PLAA (Z0041, CN, n=26)
KMT2D (Z0171, LoF, n=12)
SMARCB1 (Z0109, silencing, n=17)
SMARCB1 + CN covariates (n=15 of 732)
NF1 (Z0052, LoF, n=40)
NF2 (Z0333, LoF, n=26)
KEAP1 (Z0124, LoF, n=13)
BRCA1 (Z0205, LoF, n=14)
RB1 (Z0200, LoF, n=60)
TP53 (Z0158, LoF, n=605)
MED12 (Z0014, LoF, n=12)
VHL (Z0145, LoF, n=16)
FOCAD (Z0286, CN, n=46)
residual sensitisation
detectable, within margin (E2)
neutral
inconclusive
matched null 2.5–97.5%
BH discovery (q < 0.05)
TOST margin ±0.1
−0.1 0.0 0.1 0.2 0.3 0.4
κ_N (Chronos; 90% CI) < 0 collapse · > 0 residual sensitisation
b
Paralog axis (in-complex paralogs P)
MAP2K4 → MAP2K7 (Z0164, n=16)
KMT2D → KMT2C (Z0171, n=12)
ARID1A → ARID1B (Z0109, n=37)
EP300 → CREBBP (Z0186, n=12)
not evaluable (counted test)
inconclusive
paralog compensation
paralog compensation
−1.5 −1.0 −0.5 0.0 0.5
β_P (Chronos; 90% CI) < 0 = paralog more essential
c
Member composition of the headline-eligible events (▼ most shifted N member)
lost gene
in-complex paralog (P)
counted member (N')
no baseline direction (|b| < 0.1)
ARID1A loss Z0109 (LoF)
ARID1A ARID1B DPF2 SMARCA4 SMARCB1 SMARCC1 SMARCC2 SMARCD2 SMARCE1 IER5L SS18
β_P -0.37 (95% CI -0.51 to -0.24), null -0.07…+0.08; P−N contrast -0.39 (p 2e-08); 37 loss lines
complex-wide κ_N -0.034 (90% CI -0.072 to +0.004): neutral
EP300 loss Z0186 (LoF)
EP300 CREBBP SMARCD1 BRD9 SIRT1
β_P -1.09 (95% CI -1.46 to -0.72), null -0.10…+0.10; P−N contrast -1.04 (p 0.0002); 12 loss lines
complex-wide axis not evaluable (|N'| = 1)
FOCAD loss Z0286 (CN)
FOCAD AVEN DIS3L HBS1L PELO RNF14 SKIC2 SKIC3 VDAC1 TUT7
κ_N +0.140 (95% CI +0.085 to +0.194), null -0.062…+0.040; 62% of N' aligned > 0; 30 loss lines
Gene-level control readouts on A (own contexts, Table S1): PC6 ARID1B in LoF:ARID1A: β -0.38 (one-sided p 8e-07); PC10 PELO in FOCAD_del_or_MSI_H: β -0.66 (one-sided p 1e-06); PC10 HBS1L in FOCAD_del_or_MSI_H: β -0.21 (one-sided p 0.0002)
d
H1 (24 events)
paralog compensation (paralog axis) 2
residual sensitisation 1
collapse 0
detectable, within margin (E2) 2
neutral 11
inconclusive 6
inconclusive (null unavailable) 1
not evaluable (counted test) 1
not evaluable (|N'| < 2) 2
0 2 4 6 8 10 12
Events
e
H1S (317 events, exploratory)
paralog compensation (paralog axis) 0
residual sensitisation 0
collapse 20
detectable, within margin (E2) 4
neutral 191
inconclusive 45
inconclusive (null unavailable) 13
not evaluable (counted test) 27
not evaluable (|N'| < 2) 17
0 50 100 150 200
Events